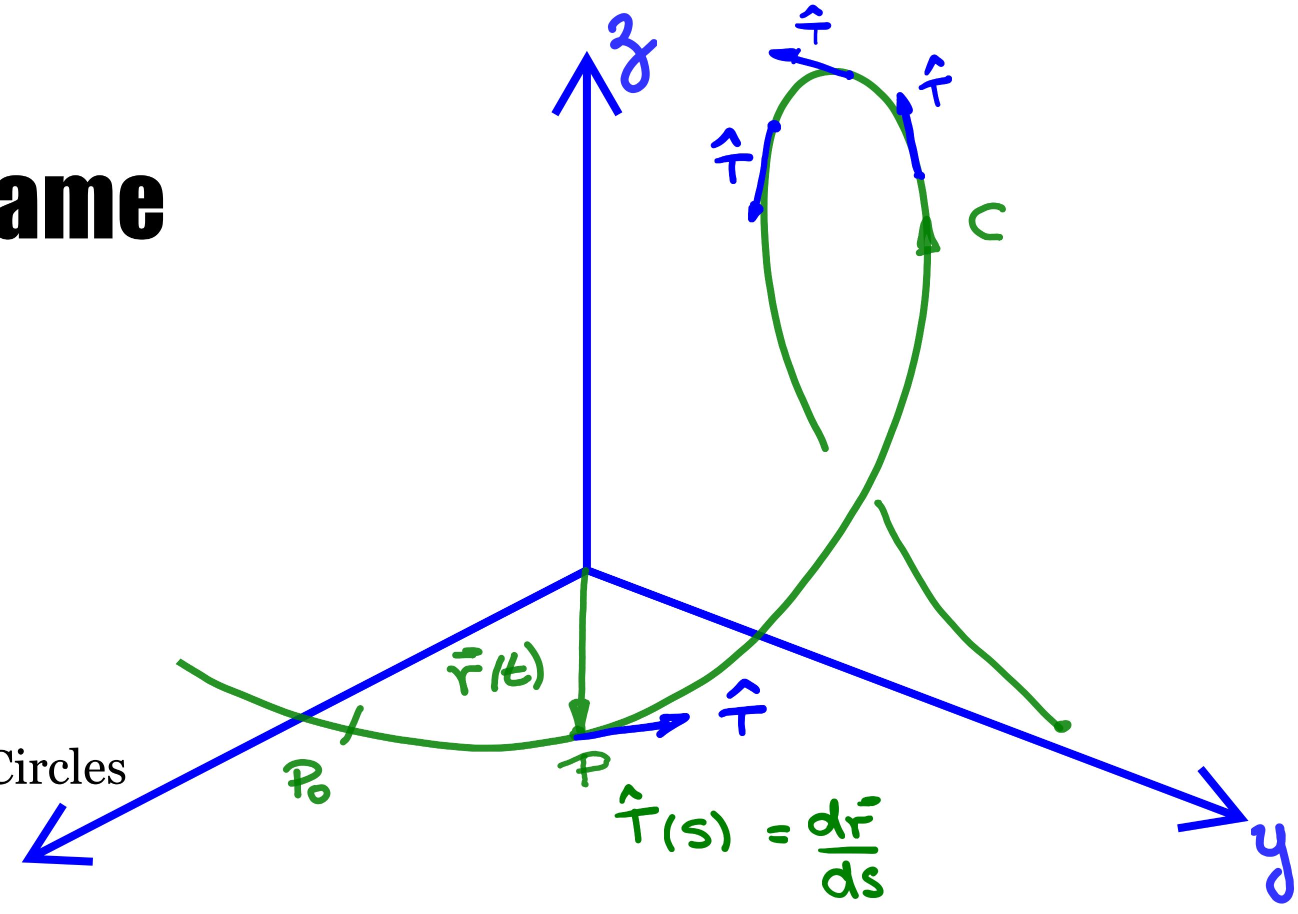
The Frenet Frame

Trim Ch 11, S.11.12



$$\hat{T}(s) = \frac{d\vec{r}}{ds}$$

Outline:

1. The Frenet Frame
2. Related Planes and Circles

CURVATURE:

$$\kappa = \left|\frac{d\hat{T}(s)}{ds}\right| \rightarrow \kappa = \frac{1}{|\vec{v}(t)|}\left|\frac{d\hat{T}(t)}{dt}\right|$$

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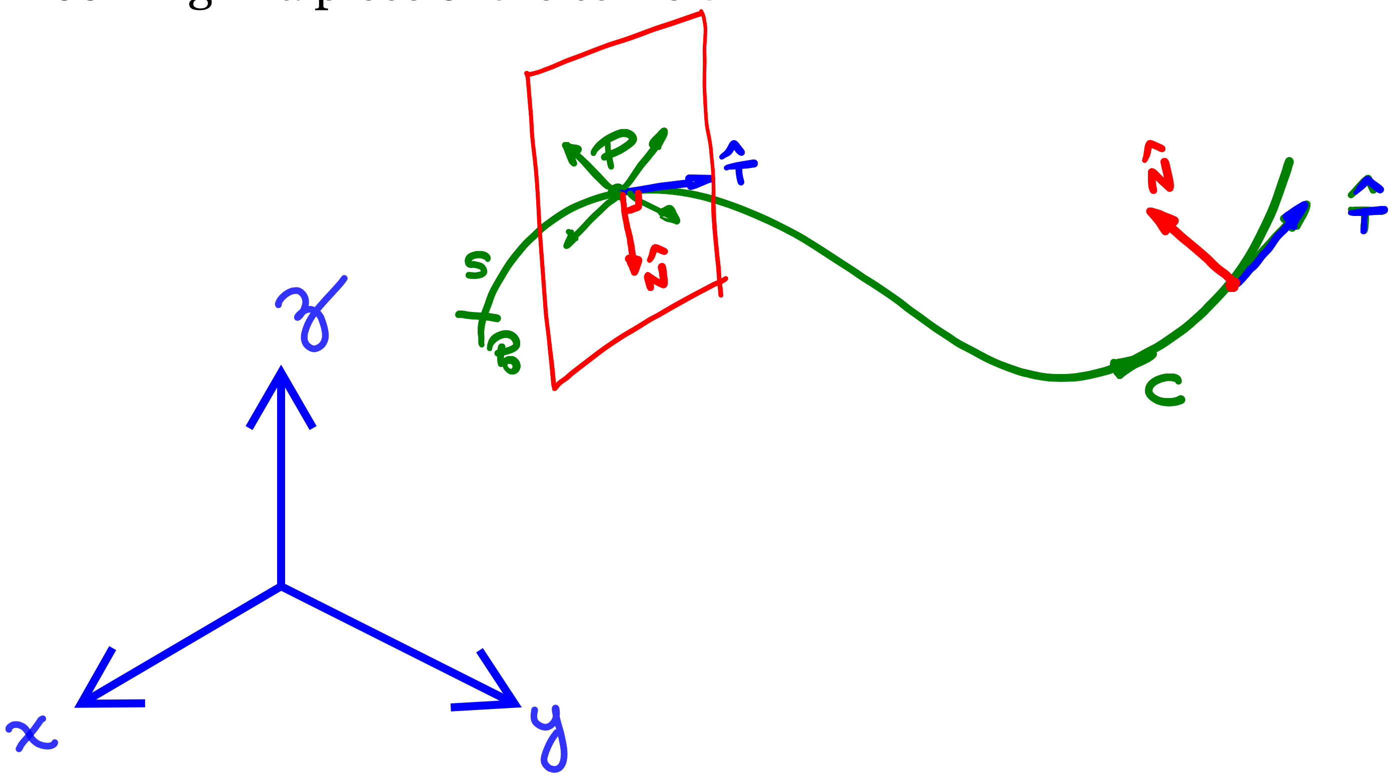
Principal Unit Normal Vector

Zooming in a piece of the curve C



The principal unit normal vector $\widehat{N}$ is one of those vectors orthogonal to $\widehat{T}$ but points in the direction in which the curve is turning (concave side of the curve).

Principal Unit Normal Vector

Let us relate the principal unit normal vector $\hat{N}$ to $\hat{T}(s)$

$$|\hat{T}(s)| = 1 \quad \rightarrow \quad \hat{T}(s) \cdot \hat{T}(s) = 1$$

$$(\hat{T}(s))^2 = 1$$

DIFFERENTIATING WITH RESPECT TO s,

$$2\underbrace{(\hat{T}(s))}_{\text{UNIT TANGENT VECTOR}} \cdot \underbrace{\frac{d\hat{T}(s)}{ds}}_{\text{NORMAL TO } \hat{T}(s)} = 0$$

$$\hat{N} = \frac{\frac{d\hat{T}}{ds}}{\left|\frac{d\hat{T}}{ds}\right|} \xrightarrow[\kappa = \left|\frac{d\hat{T}}{ds}\right|]{\text{BUT}} \rightarrow \boxed{\hat{N} = \frac{1}{\kappa}\frac{d\hat{T}}{ds}}$$

$$\rho = \frac{1}{\kappa}$$ RADIUS OF CURVATURE

Principal Unit Normal Vector

The principal unit normal vector $\widehat{N}$ for a curve given in terms of the parameter "t" can be calculated as,

$$\hat{N} = \frac{\frac{d\hat{T}}{ds}}{\left|\frac{d\hat{T}}{ds}\right|} = \frac{\frac{d\hat{T}}{dt} \cdot \cancel{\frac{dt}{ds}}}{\left|\frac{d\hat{T}}{dt}\right| \cdot \left|\cancel{\frac{dt}{ds}}\right|}$$

$$\hat{N} = \frac{\frac{d\hat{T}}{dt}}{\left|\frac{d\hat{T}}{dt}\right|}$$

Note that this expression of $\widehat{N}$ does not need parametrization as a function of s, and it does not require the value of the curvature!

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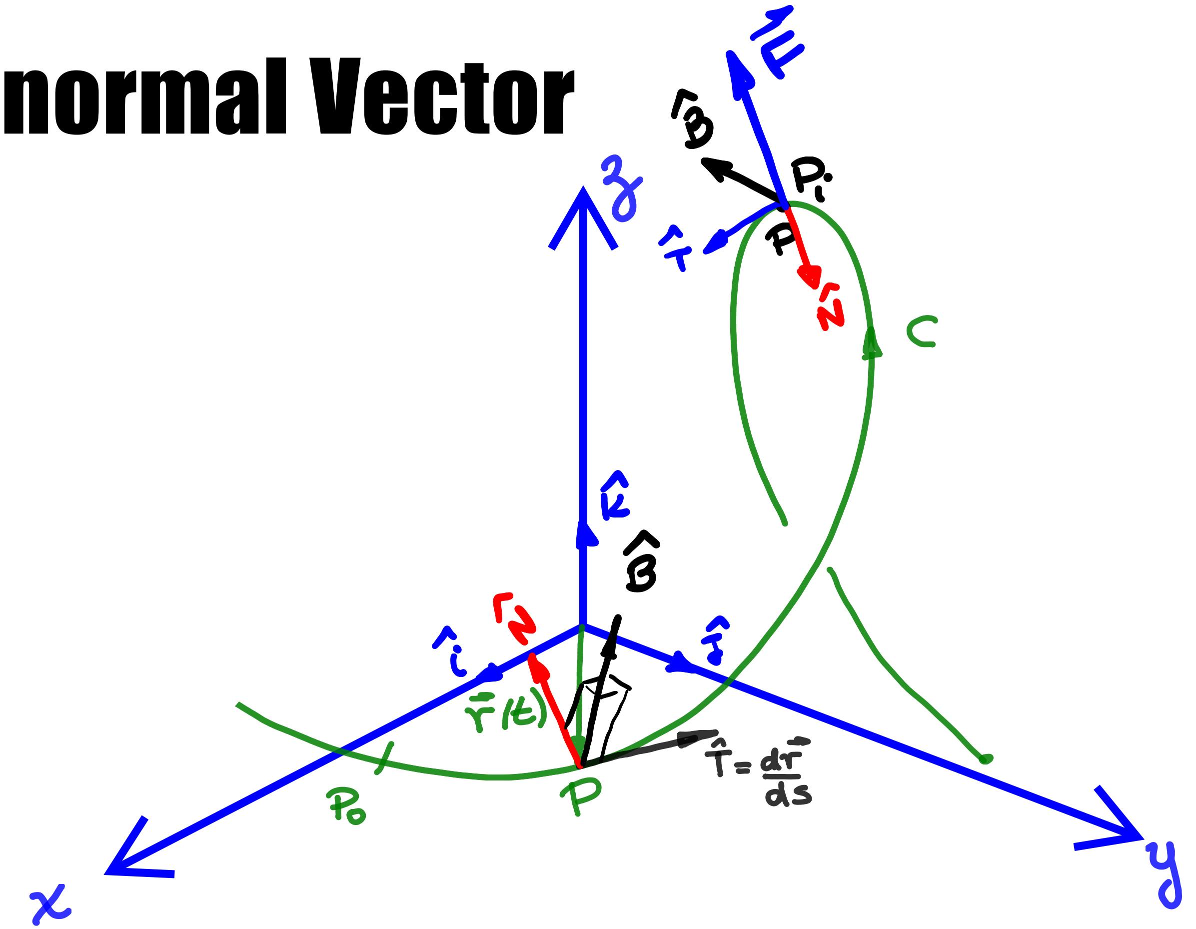
Unit Binormal Vector



The tendency of the motion to “twist” out of plane created by $\hat{T}$ and $\widehat{N}$ in the direction perpendicular to it is define by,

$$\hat{B} = \hat{T} \times \widehat{N}$$

and $\hat{B}$ is the Binormal Vector.

The Frenet Frame

- Together $\hat{T}$ and $\hat{N}$ and $\hat{B}$ define a moving right-handed vector frame called The Frenet (Fre-Nay) Frame.
- This vector frame plays a significant role in calculating paths of particles moving trough space
  - The Frenet Frame helps robots and self-driving cars **think like drivers**, focusing on the road ahead and how to move along it, rather than worrying about the whole map (like GPS coordinates).

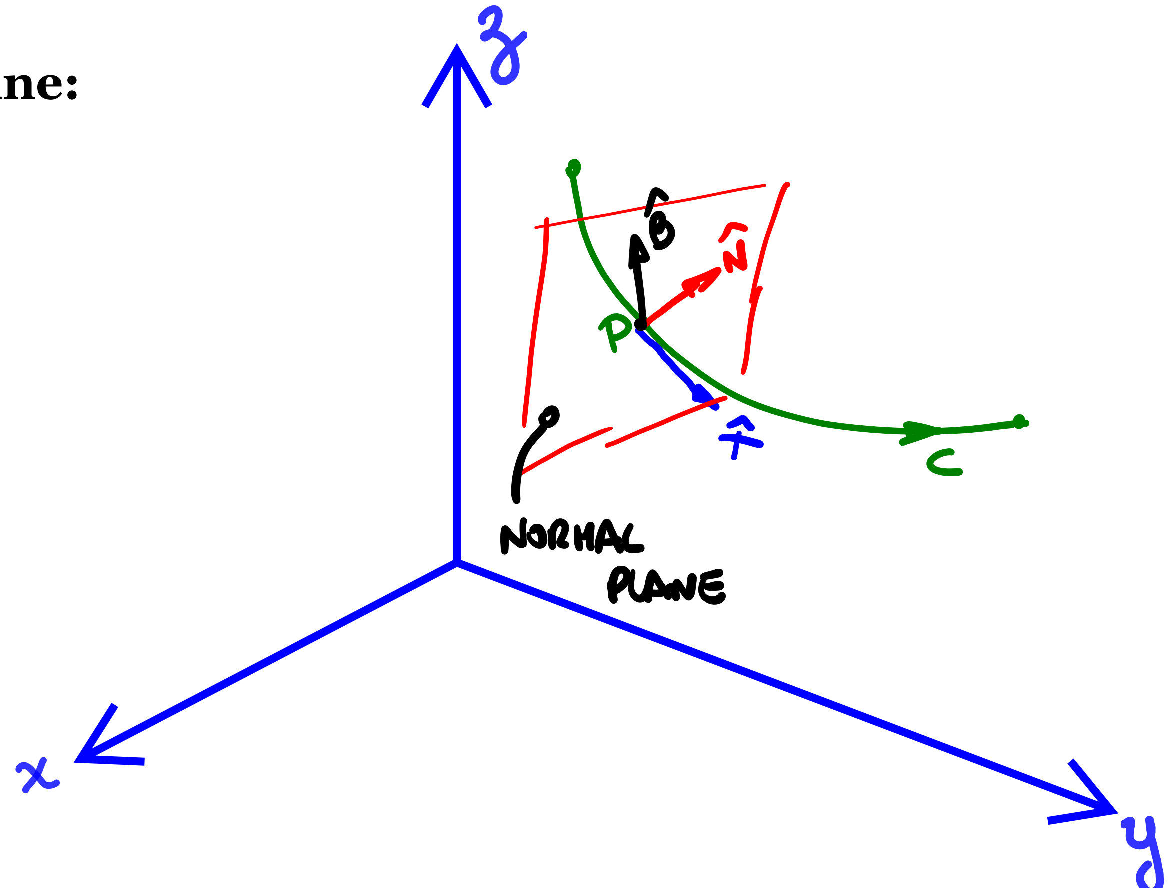
Example:

Express a vector $\vec{F}$ in terms of the Frenet Frame?

$$\vec{F} = \underbrace{\vec{F} \cdot \hat{T}}_{F_T} \hat{T} + \underbrace{\vec{F} \cdot \hat{N}}_{F_N} \hat{N} + \underbrace{\vec{F} \cdot \hat{B}}_{F_B} \hat{B}$$

Important Related Planes and Circles

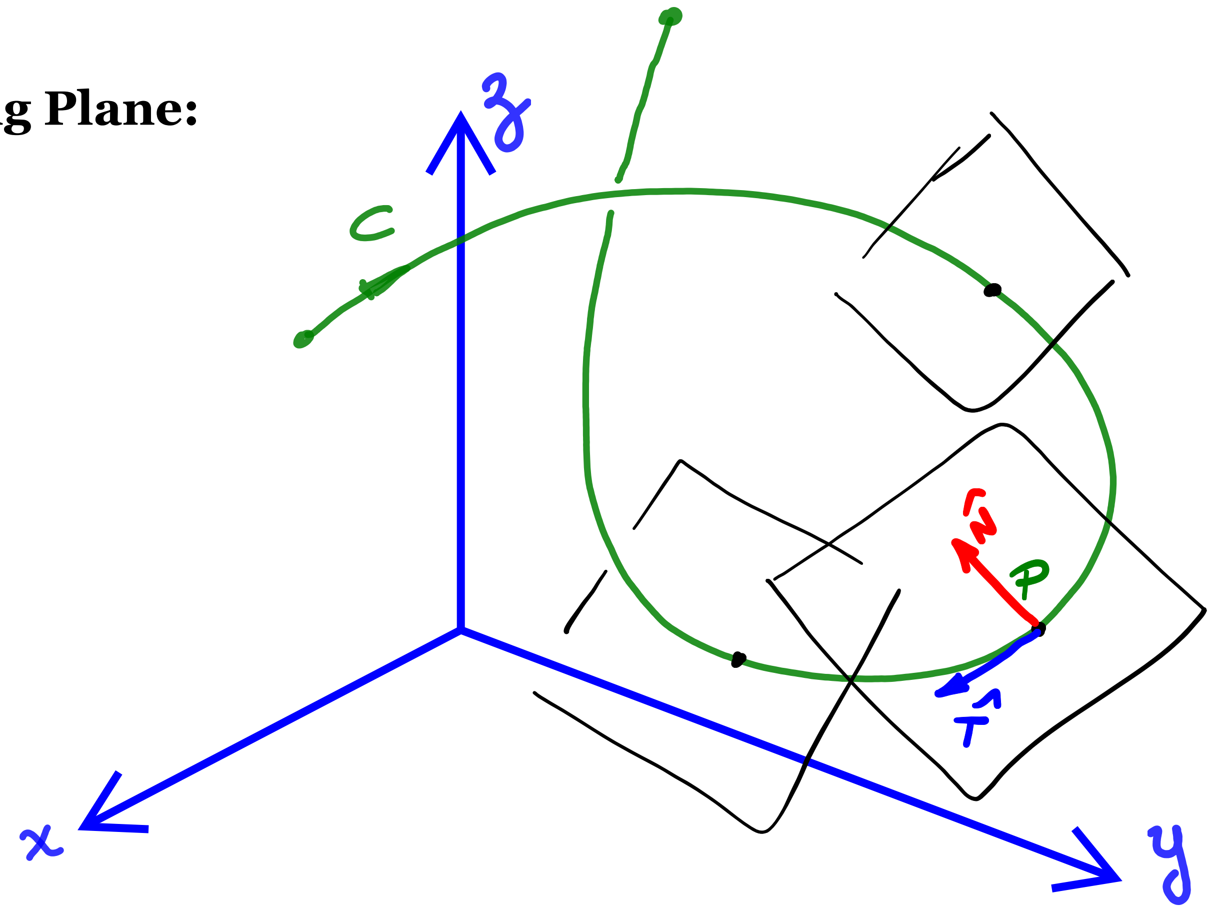
Normal Plane:



The **normal plane** is that plane that **is determined by the vectors $\hat{N}$ and $\hat{B}$** at a point P on a curve C.

Important Related Planes and Circles

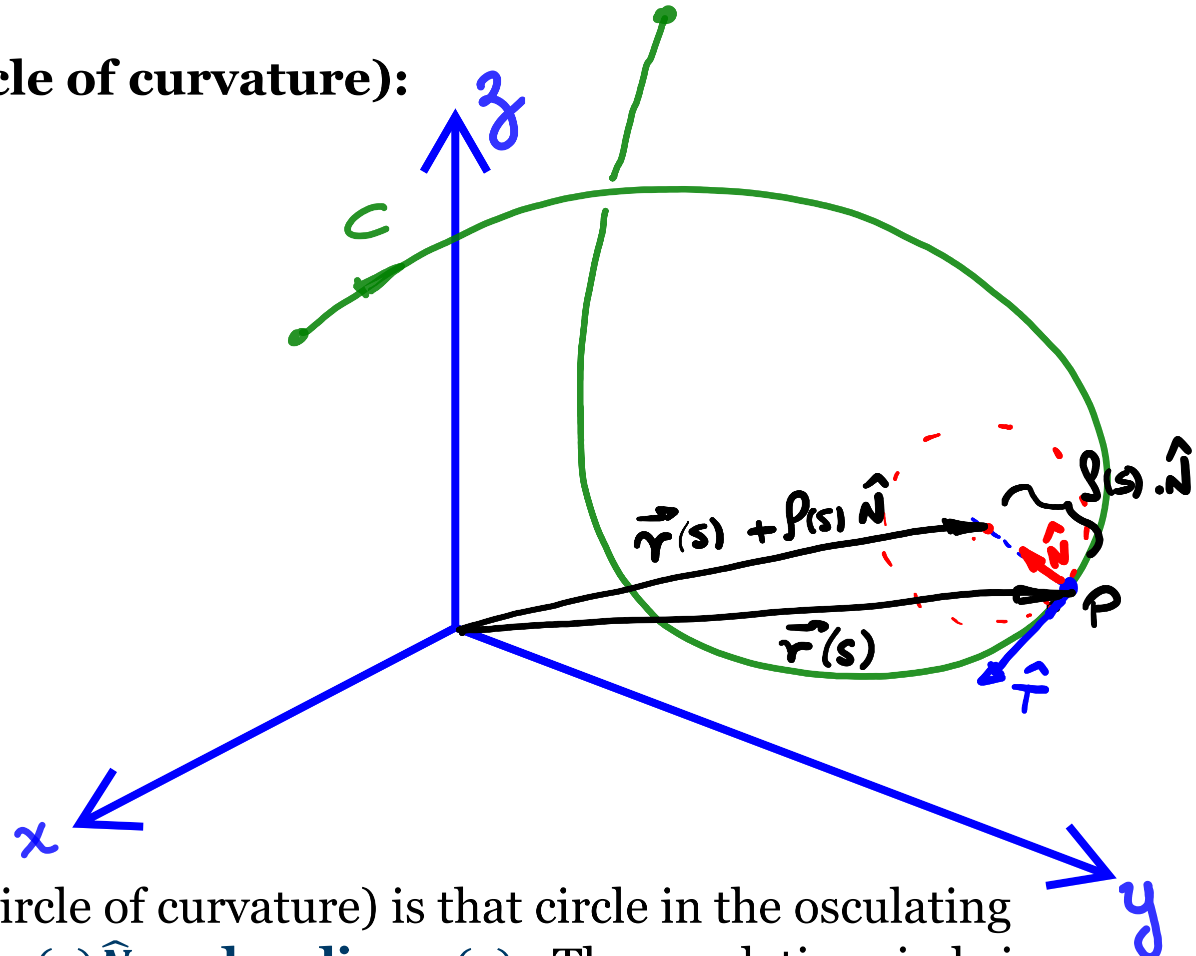
Osculating Plane:



The **osculating plane** is that plane that **is determined by the vectors $\hat{T}$ and $\hat{N}$** at a point P on a curve C.

Important Related Planes and Circles

Osculating Circle (or circle of curvature):



The **osculating circle** (or circle of curvature) is that circle in the osculating plane with **center at $\vec{r}(s) + \rho(s)\hat{N}$ and radius $\rho(s)$**. The osculating circle is the best circle that approximates the curve at P.

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Summary

- The principal unit vector $\widehat{N}$ points toward the concave side of the curve
- The tendency of the motion to "twist" out of plane created by $\hat{T}$ and $\widehat{N}$ in the direction perpendicular to it is define by $\hat{B}$
- Together $\hat{T}$ and $\widehat{N}$ and $\hat{B}$ define a moving right-handed vector frame called The Frenet (Fre-Nay) Frame, which simplifies calculations during path planning.
- Important related planes and circles for modeling the path of a particle are the normal plane, the osculating plane and the osculating circle.

Thank you